

Answer: C

**19**  $I \propto A^2$  where  $I$  is intensity and  $A$  is amplitude

$$I' = \frac{2.5}{4\pi(1.0)^2} = 0.1989$$

$$1.0 \propto (4.0 \times 10^{-6})^2 \quad \underline{\hspace{2cm}} \quad (1)$$

$$0.1989 \propto (A')^2 \quad \underline{\hspace{2cm}} \quad (2)$$

$$\frac{(2)}{(1)}, \quad A' = 1.8 \mu\text{m}$$

Answer: C